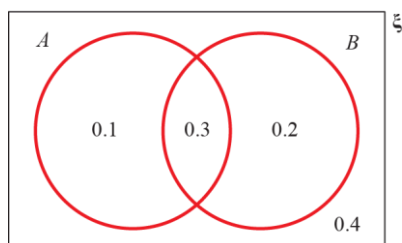


Conditional probability 2D

- 1 a Rewrite the addition formula to obtain

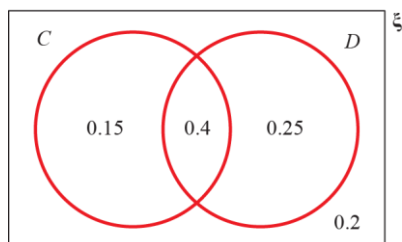
$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.4 + 0.5 - 0.6 = 0.3$$

Use this result to complete a Venn diagram to help answer the remaining parts of the question.



- b $P(A') = 0.2 + 0.4 = 0.6$
- c $P(A \cup B') = 0.4 + 0.4 = 0.8$
- d $P(A' \cup B) = 0.5 + 0.4 = 0.9$
- 2 a $P(C \cup D) = P(C) + P(D) - P(C \cap D) = 0.55 + 0.65 - 0.4 = 0.8$

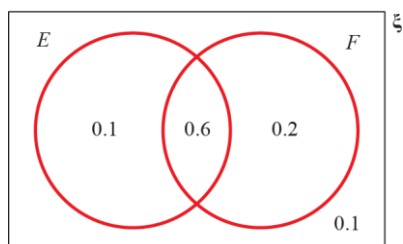
b



- i The required region is the part 'outside' of C and D , which can be found since all of the probabilities must sum to 1.
- $$P(C' \cap D') = 1 - P(C \cup D) = 1 - 0.8 = 0.2$$
- ii $P(C|D) = \frac{P(C \cap D)}{P(D)} = \frac{0.4}{0.65} = 0.615 \text{ (3 s.f.)}$
- iii $P(C|D') = \frac{P(C \cap D')}{P(D')} = \frac{0.15}{0.35} = \frac{3}{7} = 0.429 \text{ (3 s.f.)}$
- c From part **iii**, it is known that $P(C|D) \neq P(C)$ so the two events are not independent. Alternatively, show that $P(C) \times P(D) \neq P(C \cap D)$.

3 a $P(E \cup F) = P(E) + P(F) - P(E \cap F) = 0.7 + 0.8 - 0.6 = 0.9$

b



- i** The required region is within E as well as everything outside F . It includes three of the four regions in the Venn diagram.

$$P(E \cup F') = 0.1 + 0.6 + 0.1 = 0.8$$

- ii** The required region is that part of F that does not intersect E .

$$P(E \cap F') = 0.2$$

iii $P(E|F') = \frac{P(E \cap F')}{P(F')} = \frac{0.1}{0.1 + 0.1} = \frac{1}{2} = 0.5$

4 a $P(T \cup Q) = P(T) + P(Q) - P(T \cap Q)$

$$0.75 = 3P(T \cap Q) + 3P(T \cap Q) - P(T \cap Q)$$

$$5P(T \cap Q) = 0.75$$

$$P(T \cap Q) = 0.15$$

- b** As $P(T) = P(Q)$, using $P(T \cup Q) = P(T) + P(Q) - P(T \cap Q)$ gives

$$0.75 = 2P(T) - 0.15$$

$$\Rightarrow 2P(T) = 0.9$$

$$\Rightarrow P(T) = 0.45$$

c $P(Q') = 1 - P(Q) = 1 - P(T) = 1 - 0.45 = 0.55$

d $P(T' \cap Q') = 1 - P(T \cup Q) = 1 - 0.75 = 0.25$

e $P(T \cap Q') = P(T) - P(T \cap Q) = 0.45 - 0.15 = 0.3$

- 5** Let F be the event has a freezer and D be the event has a dishwasher. The question requires finding $P(F \cap D)$. Use the addition formula

$$P(F \cap D) = P(F) + P(D) - P(F \cup D) = 0.7 + 0.2 - 0.8 = 0.1$$

- 6 a** Use the multiplication formula for conditional probability to find $P(A \cap B)$
 $P(A \cap B) = P(A|B) \times P(B) = 0.4 \times 0.5 = 0.2$

Now use the multiplication formula again to find $P(B|A)$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{0.2}{0.4} = \frac{1}{2} = 0.5$$

- b** Use the addition formula to find $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.7$$

Now $P(A' \cap B')$ can be found as it is the region outside $P(A \cup B)$

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.7 = 0.3$$

- c** $P(A' \cap B) = P(B) - P(A \cap B) = 0.5 - 0.2 = 0.3$

- 7 a** First use the addition formula to find $P(A \cap B)$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = \frac{1}{4} + \frac{1}{2} - \frac{3}{5} = \frac{3}{20}$$

Now use the multiplication formula to

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{3}{20}}{\frac{1}{2}} = \frac{3}{10} = 0.3$$

- b** $P(A' \cap B) = P(B) - P(A \cap B) = \frac{1}{2} - \frac{3}{20} = \frac{7}{20} = 0.35$

- c** $P(A' \cap B') = 1 - P(A \cup B) = 1 - \frac{3}{5} = \frac{2}{5} = 0.4$

- 8 a** $P(C \cap D) = P(C|D) \times P(D) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12} = 0.0833$ (3 s.f.)

- b** $P(C \cap D') = P(C|D') \times P(D') = \frac{1}{5} \times \left(1 - \frac{1}{4}\right) = \frac{1}{5} \times \frac{3}{4} = \frac{3}{20} = 0.15$

- c** $P(C) = P(C \cap D') + P(C \cap D) = \frac{3}{20} + \frac{1}{12} = \frac{9}{60} + \frac{5}{60} = \frac{14}{60} = \frac{7}{30} = 0.233$ (3 s.f.)

- d** $P(D|C) = \frac{P(D \cap C)}{P(C)} = \frac{\frac{1}{12}}{\frac{7}{30}} = \frac{30}{84} = \frac{5}{14} = 0.357$ (3 s.f.)

- e** $P(D'|C) = 1 - P(D|C) = 1 - \frac{5}{14} = \frac{9}{14} = 0.643$ (3 s.f.)

- 8 f** $P(D'|C') = \frac{P(C' \cap D')}{P(C')} = \frac{1 - P(C \cup D)}{P(C')}$

$$\text{However, } P(C') = 1 - P(C) = \frac{7}{30} = \frac{23}{30}$$

$$\text{And } P(C \cup D) = P(C) + P(D) - P(C \cap D) = \frac{7}{30} + \frac{1}{4} - \frac{1}{12} = \frac{24}{60} = \frac{2}{5}$$

$$\text{So } P(D'|C') = \frac{P(C' \cap D')}{P(C')} = \frac{1 - P(C \cup D)}{P(C')} = \frac{1 - \frac{2}{5}}{\frac{23}{30}} = \frac{\frac{3}{5}}{\frac{23}{30}} = \frac{3}{5} \times \frac{30}{23} = \frac{18}{23} = 0.783 \text{ (3 s.f.)}$$

9 a $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.42 + 0.37 - 0.12 = 0.67$

b $P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{P(A) - P(A \cap B)}{1 - P(B)} = \frac{0.42 - 0.12}{1 - 0.37} = \frac{0.3}{0.63} = 0.476 \text{ (3 s.f.)}$

c Since the events A and C are independent, $P(A \cap C) = P(A) \times P(C) = 0.42 \times 0.3 = 0.126$

d Since B and C are mutually exclusive, there is no need to have an intersection between B and C on the diagram. Work out the probabilities associated with each region as follows:

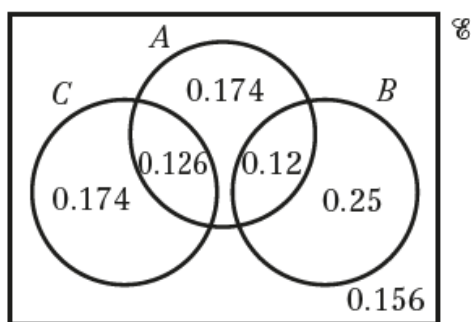
$$P(C \cap A') = P(C) - P(A \cap C) = 0.3 - 0.126 = 0.174$$

$$P(B \cap A') = P(B) - P(A \cap B) = 0.37 - 0.12 = 0.25$$

$$P(A \cap B' \cap C') = P(A) - P(A \cap B) - P(A \cap C) = 0.42 - 0.12 - 0.126 = 0.174$$

$$P(A \cup B \cup C) = 0.174 + 0.126 + 0.174 + 0.12 + 0.25 = 0.844$$

$$P(A' \cap B' \cap C') = 1 - P(A \cup B \cup C) = 1 - 0.844 = 0.156$$



e $P((A' \cup C)') = 1 - P(A' \cup C)$

Use the Venn diagram to find $P(A' \cup C) = 0.174 + 0.126 + 0.25 + 0.156 = 0.706$

So $P((A' \cup C)') = 1 - 0.706 = 0.294$

10 a B and C are independent: $P(B \cap C) = P(B) \times P(C) = 0.7 \times 0.4 = 0.28$

b Using part **a**, $P(B|C) = \frac{P(B \cap C)}{P(C)} = \frac{0.28}{0.4} = \frac{7}{10} = 0.7$

c $P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{P(A) - P(A \cap B)}{1 - P(B)} = \frac{0.4 - 0.3}{1 - 0.7} = \frac{0.1}{0.3} = 0.333$ (3 s.f.)

d $P((B \cap C)|A') = \frac{P((B \cap C) \cap A')}{P(A')} = \frac{P(B \cap C) - P(A \cap B \cap C)}{1 - P(A)}$

As A and C are mutually exclusive, $P(A \cap B \cap C) = 0$

So $P((B \cap C)|A') = \frac{P(B \cap C)}{1 - P(A)} = \frac{0.28}{1 - 0.4} = \frac{0.28}{0.6} = 0.467$ (3 s.f.)

11 a This requires finding $P(A \cap B)$

First find $P(A \cup B)$

$P(A \cup B) = 0.9$ as $P(A \cup B) + P(A' \cap B') = 1$

Using the addition rule gives

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.3 + 0.7 - 0.9 = 0.1$

b This requires finding $P(A|B)$

$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.1}{0.7} = 0.143$ (3 s.f.)

c Test whether the events are independent

$P(A) \times P(B) = 0.3 \times 0.7 = 0.21$, $P(A \cap B) = 0.1$

So the events are not independent. If Anna is late, Bella is *less* likely to be late and vice versa.

12 a The probability that both John and Kayleigh win their matches is $P(J \cap K)$

$P(J \cap K) = P(J) + P(K) - P(J \cup K) = 0.6 + 0.7 - 0.8 = 0.5$

b $P(J|K') = \frac{P(J \cap K')}{P(K')} = \frac{P(J) - P(J \cap K)}{1 - P(K)} = \frac{0.6 - 0.5}{1 - 0.7} = \frac{0.1}{0.3} = 0.333$ (3 s.f.)

c $P(K|J) = \frac{P(J \cap K)}{P(J)} = \frac{0.5}{0.6} = 0.833$ (3 s.f.)

d $P(K|J) = 0.833$ (3 s.f.), $P(K) = 0.7$, so $P(K|J) \neq P(K)$ $P(K|J) = 0.833 \dots \neq P(K) = 0.7$. So J and K are not independent.

Challenge

a The probability function must sum to 1. Therefore $k + 2k + 3k + 4k + 5k = 1 \Rightarrow 15k = 1 \Rightarrow k = \frac{1}{15}$

$$\mathbf{b} \quad P(X = 5 | X > 2) = \frac{P(X = 5)}{P(X > 2)} = \frac{\frac{5}{15}}{1 - P(X = 1 \cup X = 2)} = \frac{\frac{5}{15}}{1 - \frac{3}{15}} = \frac{\frac{5}{15}}{\frac{12}{15}} = \frac{5}{12}$$

$$\mathbf{c} \quad P(X \text{ is odd} | X \text{ is prime}) = \frac{P(X \text{ is odd and prime})}{P(X \text{ is prime})} = \frac{P(X = 3 \cup X = 5)}{P(X = 2 \cup X = 3 \cup X = 5)} = \frac{\frac{3+5}{15}}{\frac{2+3+5}{15}} = \frac{8}{10} = \frac{4}{5}$$